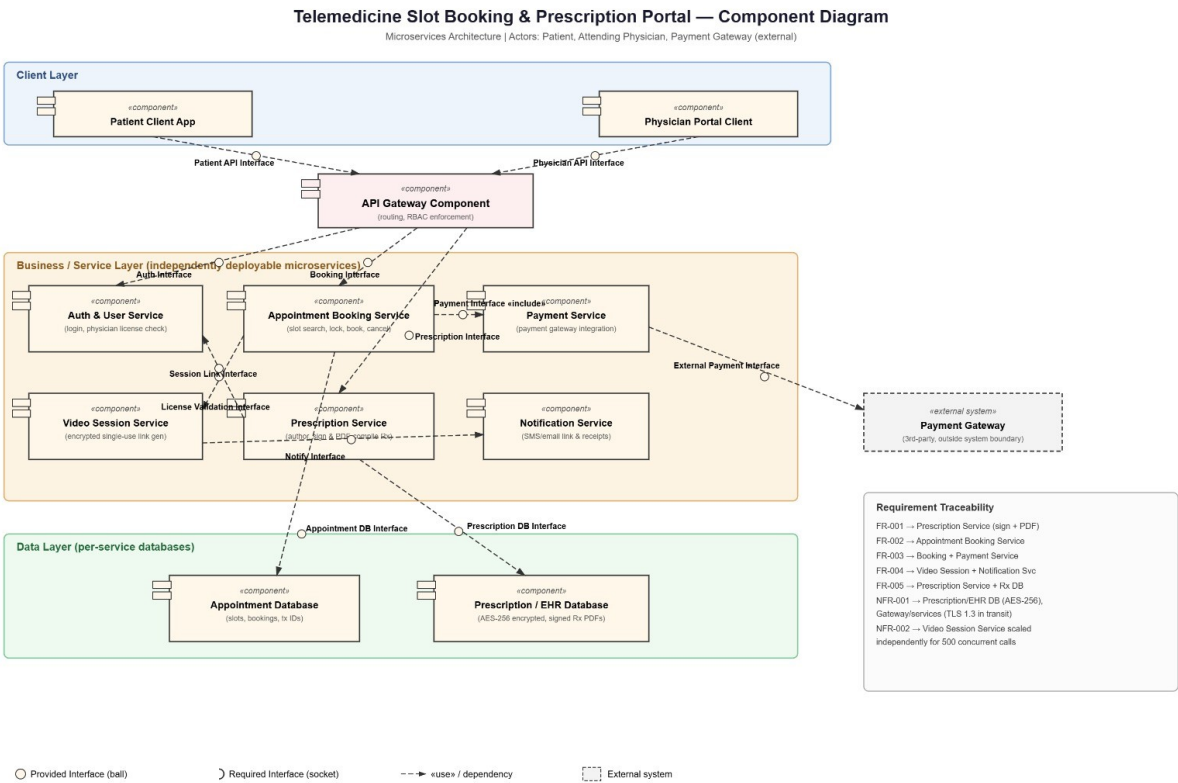


Telemedicine Slot Booking & Prescription Portal

Lab 3 – Component Diagram (Microservices Architecture)



Architectural Justification

We chose Microservices Architecture for the Telemedicine Slot Booking & Prescription Portal System.

1. Scenario Recap: Key Requirements and Challenges

The portal serves three actors — Patient, Attending Physician, and an external Payment Gateway — across five core functional flows (FR-001 to FR-005): doctor-slot discovery, concurrency-safe slot booking gated on payment, digitally signed prescription authoring, time-boxed encrypted video links, and post-consultation prescription retrieval. Two non-functional requirements shape the architecture directly: NFR-001 demands TLS 1.3 in transit and AES-256 at rest for clinical data and video metadata, and NFR-002 demands 99.5% monthly uptime with 500 concurrent video consultations at under 150ms jitter. The main challenges are therefore (a) protecting PHI and signed prescriptions, (b) preventing double-booking under concurrent load, (c) isolating a compute- and bandwidth-heavy video workload from the rest of the system, and (d) tolerating failure or slowness in the third-party Payment Gateway without stalling clinical workflows.

2. Architectural Style Analysis

Layered Architecture (Presentation / Business / Data) offers a clear separation of concerns and is simple to reason about, but every layer typically shares a single deployment and scaling unit. That means a spike in video-consultation traffic would force the entire application tier — including booking, prescriptions, and notifications — to scale together, and a bug or overload in one module (e.g., video) risks degrading unrelated ones (e.g., prescription signing). This tight coupling is a poor fit for NFR-002's isolated 500-concurrent-call target.

Client-Server Architecture centralizes logic and data on one server, which is easy to deploy and keeps data consistent, but it creates a single point of failure and a scalability bottleneck: if the server (or its single database) is saturated by video sessions or a payment gateway slowdown, booking and prescription access degrade too. It also concentrates all PHI behind one boundary, which is a weaker security posture than isolating sensitive services.

Microservices Architecture decomposes the system into small, independently deployable services, each of which can be scaled, secured, and released on its own. The operational cost is added complexity (service discovery, network latency, eventual consistency across services), but for this scenario those costs are outweighed by the ability to scale the video workload independently, isolate the Payment Gateway integration from clinical flows, and apply tighter, service-scoped access control around the Prescription/EHR data — all requirements this system explicitly has. This is why Microservices was selected over the other two styles.

3. Architectural Choice and Component Identification

The system is decomposed into six business-layer microservices — Auth & User Service, Appointment Booking Service, Payment Service, Video Session Service, Prescription Service, and Notification Service — fronted by an API Gateway that performs routing and RBAC enforcement for two clients, the Patient Client App and the Physician Portal Client. Each service owns its own data: the Appointment Booking Service persists to the Appointment Database, and the Prescription Service persists to the Prescription/EHR Database (AES-256 encrypted, storing signed Rx PDFs). The Payment Service is the sole integration point with the external Payment Gateway, reached over an External Payment Interface, keeping third-party dependency logic out of every other service.

4. Reason 1: Independent Scaling for Uneven, Workload-Specific Load

NFR-002 requires the portal to support at least 500 concurrent video consultations without quality degradation, while other functions — viewing doctor specialties, downloading a signed prescription, sending a notification — see comparatively lighter and steadier load. Because microservices are deployed independently, the Video Session Service can be horizontally scaled (more instances, dedicated bandwidth) purely in response to concurrent-call volume, without over-provisioning the Notification Service, Prescription Service, or Appointment Booking Service, which do not share that load profile. This targeted scaling is only possible because service boundaries in the diagram are also deployment boundaries.

5. Reason 2: Fault Isolation for a Multi-Actor, Multi-Vendor System

The system integrates with an external Payment Gateway (FR-003) and must keep booking, prescription, and video-session flows independently reliable (FR-002, FR-004, FR-005). In a microservices architecture, a slowdown or outage in the Payment Service — or in the external gateway it depends on — does not bring down the Appointment Booking Service, Prescription Service, or Video Session Service; each fails and recovers independently. This matters for a healthcare application specifically: a physician must still be able to author and sign a prescription (UC-07) or join a scheduled video consultation (UC-06) even if

payment processing is temporarily degraded, since those are clinically time-sensitive actions that shouldn't be coupled to a billing dependency.

6. Security Advantage

Isolating the Prescription Service and its Prescription/EHR Database as a separate microservice narrows the blast radius of a breach: patient health information and signed prescription PDFs, encrypted at rest with AES-256 (NFR-001), are confined to one service boundary with its own access controls, rather than being directly reachable from every module the way they would be inside a monolithic data layer. The API Gateway centralizes authentication and RBAC enforcement at the edge, so only authorized, role-checked requests — a patient viewing their own records, or a physician with a validated license (UC-08) authoring a prescription — ever reach the Prescription or Video Session services. All gateway-to-client and inter-service traffic runs over TLS 1.3, satisfying NFR-001's in-transit requirement without requiring every service to independently implement it from scratch.

7. Performance Benefit

Because each microservice is deployed and scaled independently, the Video Session Service can be provisioned with dedicated compute and network resources sized specifically to sustain 500 concurrent calls at under 150ms jitter (NFR-002), without that scaling requiring — or being throttled by — the Appointment Booking or Prescription services, which have very different resource profiles. This targeted, service-specific scaling avoids the blanket over-provisioning (and associated cost) that a layered or client-server deployment would need in order to protect video quality during peak booking hours, since those styles would scale the whole application tier together rather than the one component under load.